

1. Dataset

4 species, 5 studies
389,609 sites
2,334 positives
full provenance

2. Benchmark

6 models, 10 seeds
within-dataset splits
literature-comparable
claim class only

3. Freeze

weights + features +
candidates + controls
SHA256-registered
co-signed 2026-08-13

4. Calibration

mock-blind diagnostic
20 unseen tomato sites
7 Arabidopsis controls
expectations set pre-blind

5. Preregistration

SAP + analysis plan
OSF-registered DOI
primary endpoint fixed
success criteria fixed

6. Blind cohort

593 sites assayed blind
one-sided Fisher exact
reported regardless
of outcome

Binding rule — no model, feature, candidate-table, threshold, or control-set change after unblinding;
every artifact hash-verified; negative results and exclusions preserved and archived.

enrichment met + mechanism
upgraded submission target

enrichment met, no mechanism
resource and validation report

enrichment not met
negative result, first-class finding